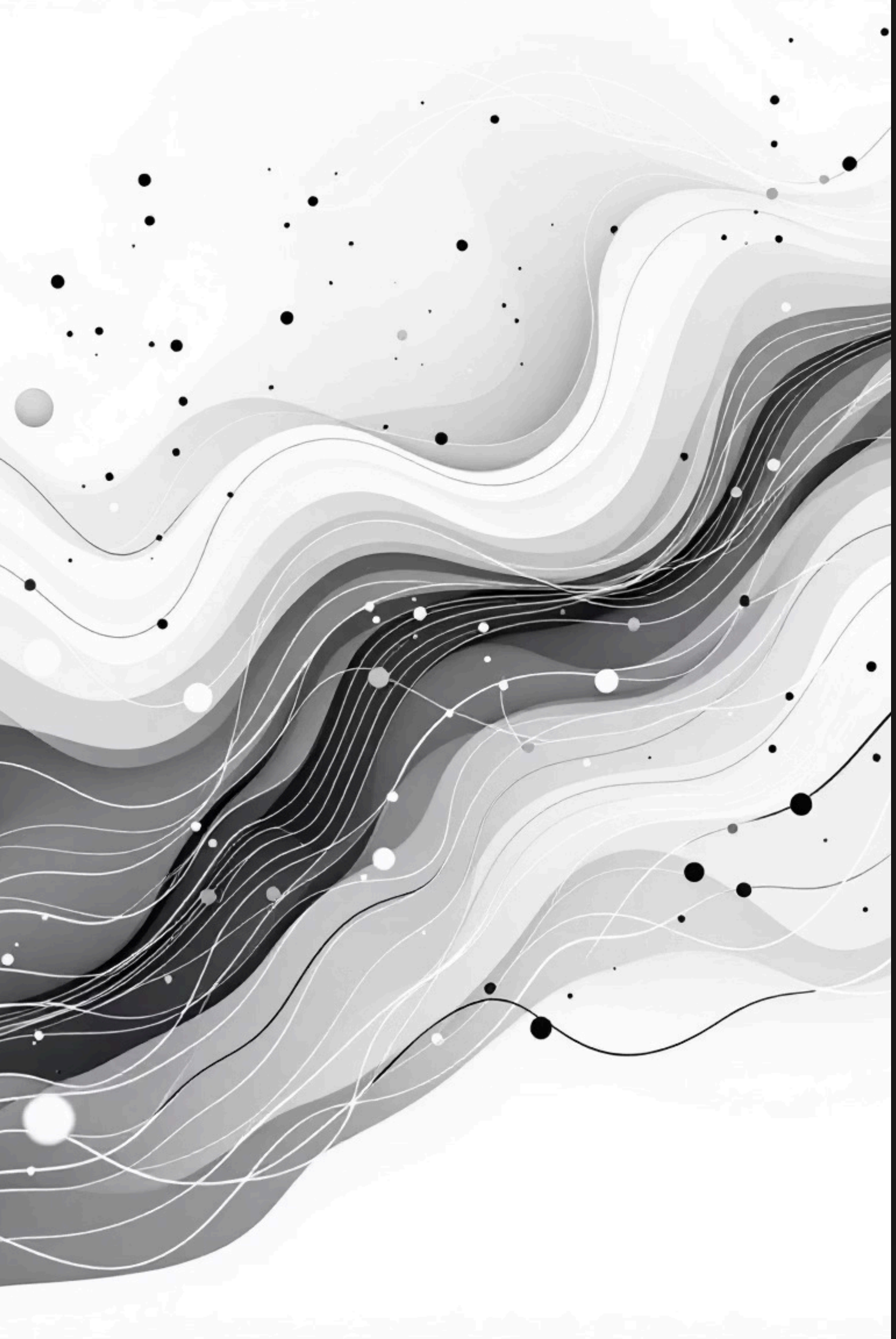


InsightFlow: Predictive Analytics and Modeling

A comprehensive machine learning project focused on heart disease prediction using advanced analytics and modeling techniques. This presentation walks through the complete data science workflow from data preparation to model evaluation and selection.

Project Overview and Setup

Core Objectives

- Load and prepare the heart disease dataset
- Build predictive machine learning models
- Train models and generate predictions
- Evaluate model performance comprehensively

Technical Environment

The project utilizes Python's essential data science libraries including pandas for data manipulation, numpy for numerical operations, matplotlib and seaborn for visualization. A random seed of 1234 ensures reproducibility across all experiments.

Dataset Source

The analysis uses an anonymized heart disease dataset containing patient medical data and diagnostic outcomes. The dataset is publicly available from the zero-to-mastery-ml GitHub repository and includes 303 patient records with 14 features.

Dataset Exploration and Initial Analysis

Dataset Dimensions

303 rows × 14 columns

Complete dataset with no missing values, ensuring data quality and reliability for modeling

Target Distribution

165 positive cases

138 negative cases

Relatively balanced classes with 54.5% positive and 45.5% negative cases

Data Quality

Zero missing values

All features have appropriate data types with no null values requiring imputation

Initial examination revealed a clean dataset with proper data types across all features. The age range spans from 29 to 77 years with a mean of 54.4 years. Key physiological measurements include resting blood pressure (trestbps) averaging 131.6 mm Hg and maximum heart rate (thalach) averaging 149.6 bpm.

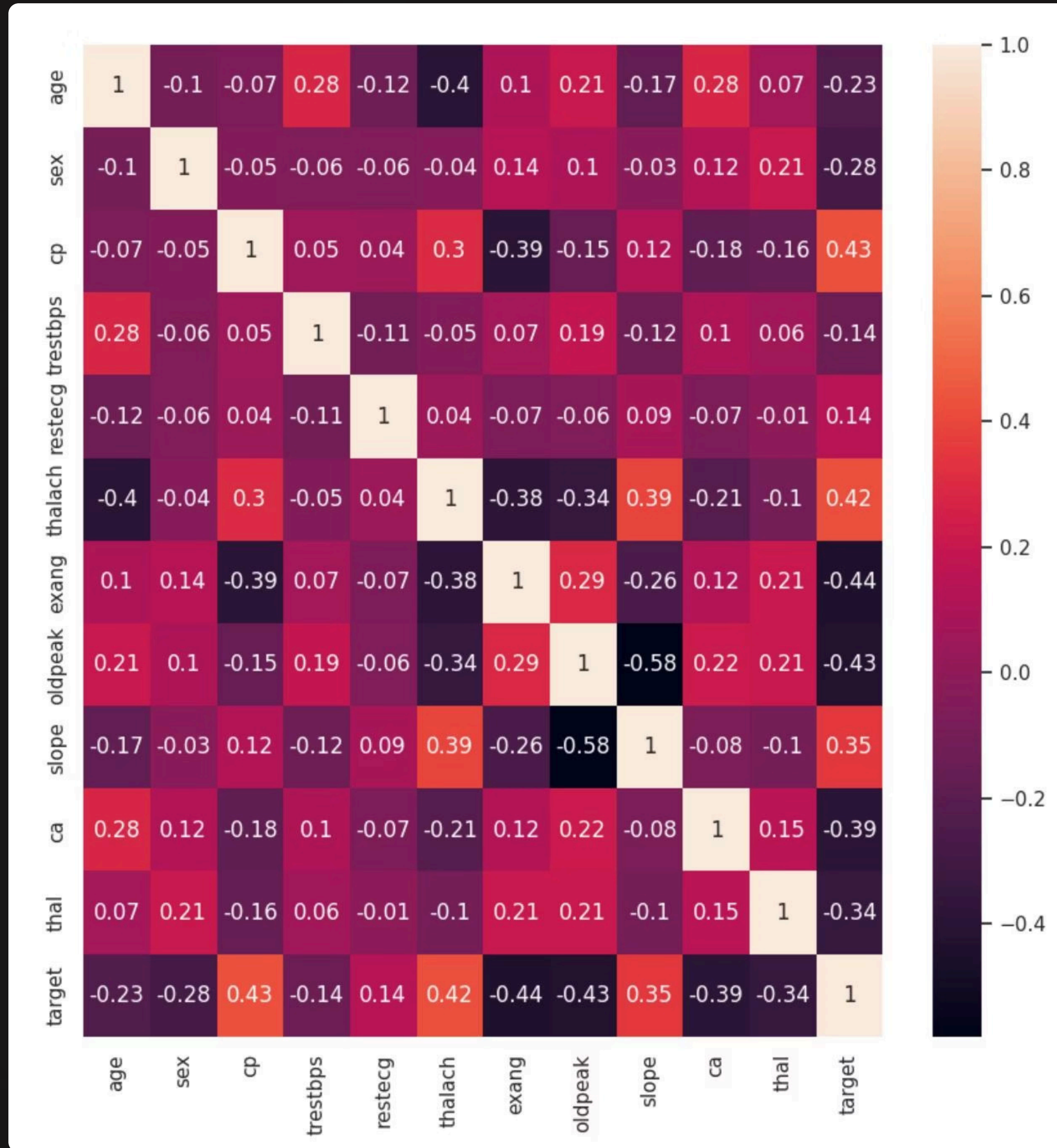
Correlation Analysis and Feature Selection

Top Correlating Features

- **cp (chest pain):** 0.434 correlation
- **thalach (max heart rate):** 0.422
- **slope:** 0.346
- **exang (exercise angina):** -0.437
- **oldpeak:** -0.431
- **ca (vessels):** -0.392

Features Removed

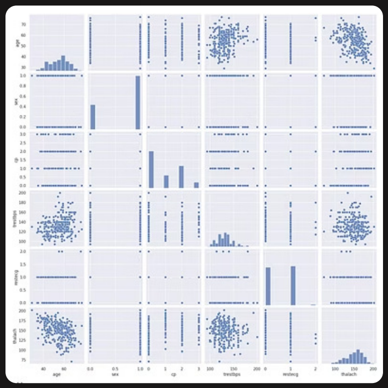
Based on correlation analysis, **fbs (fasting blood sugar)** and **chol (cholesterol)** were eliminated due to low correlation with the target variable (-0.028 and -0.085 respectively).

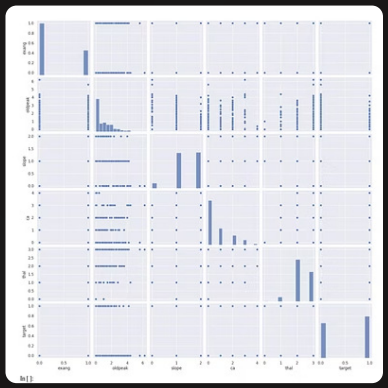


Correlation heatmap showing relationships between features and target variable

Visual Data Exploration

Comprehensive pairplot analysis revealed important patterns and relationships within the dataset. The visualizations were split into two groups to maintain clarity and readability.



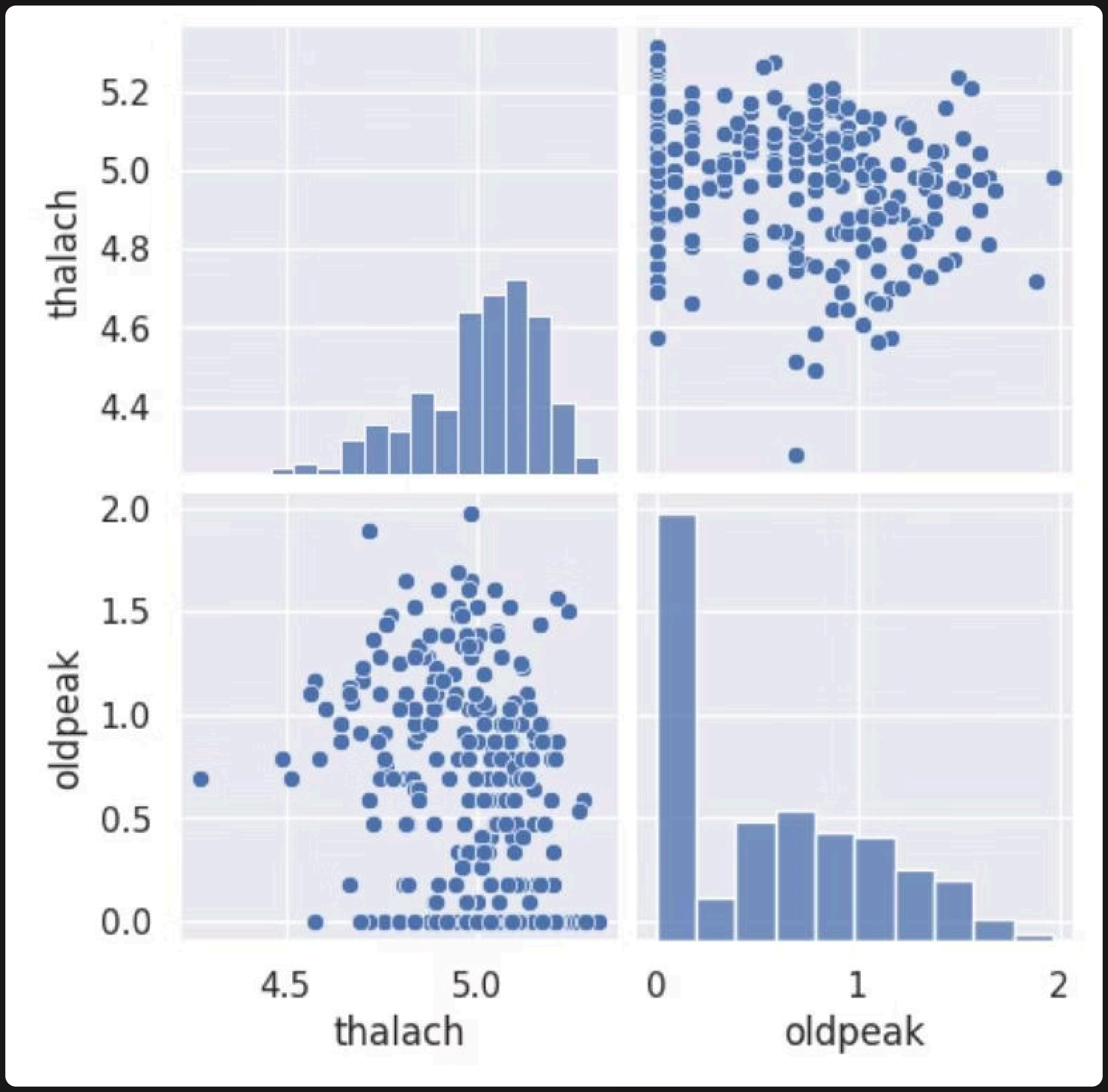


Data Transformation

Logarithmic transformation was applied to columns 5 (thalach) and 7 (oldpeak) to normalize their distributions and improve model performance. This transformation helps handle skewed data and reduces the impact of outliers.

Post-Transformation Results

After applying log transformation, the distributions became more normalized, making them more suitable for machine learning algorithms that assume normally distributed features.



Model Preparation and Training Split

01

Feature-Target Separation

Independent variables (X) were separated from the target variable (y) to prepare for supervised learning. The target column indicates presence or absence of heart disease.

02

Train-Test Split

Dataset divided into 70% training (212 samples) and 30% testing (91 samples) using stratified sampling with random_state=1234 for reproducibility.

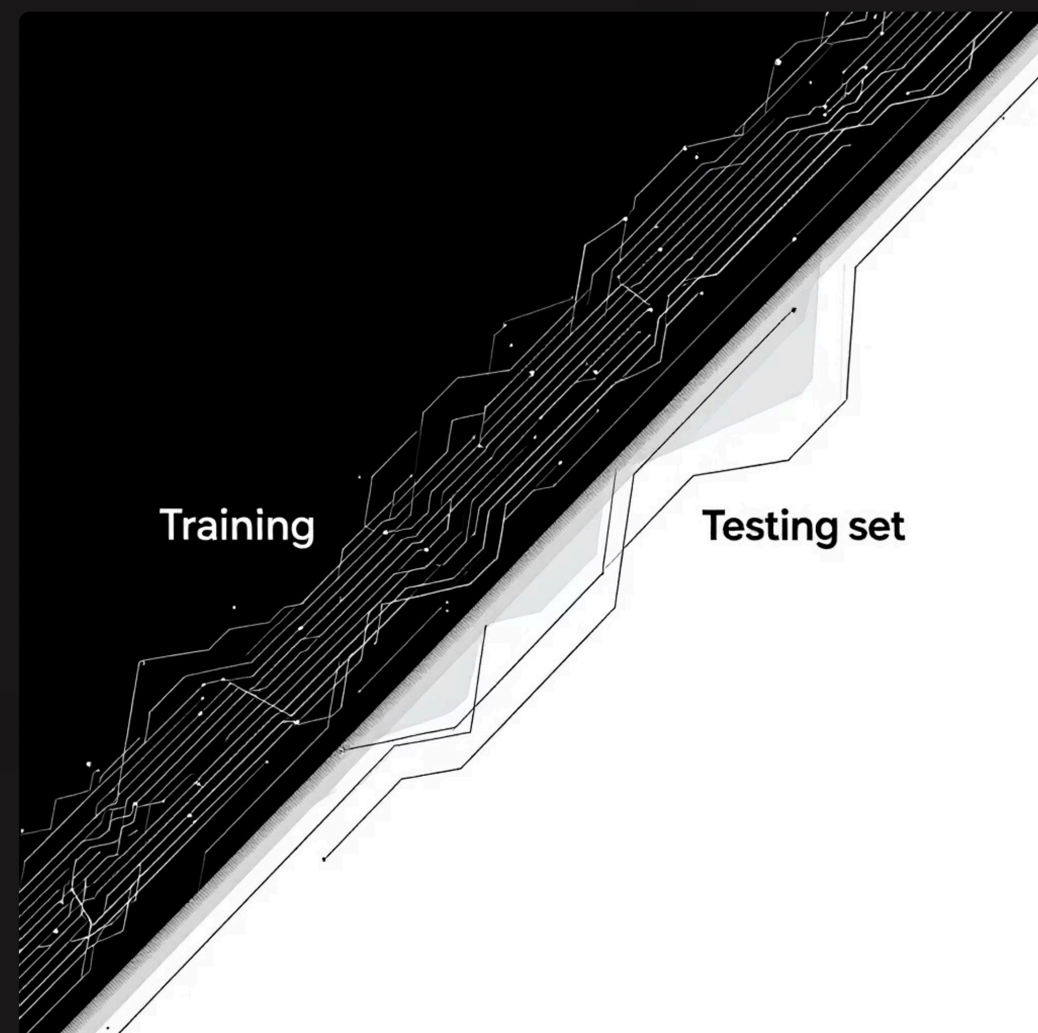
03

Feature Scaling

StandardScaler applied to normalize features, ensuring all variables contribute equally to model training. This is crucial for distance-based algorithms and improves convergence.

Final Dataset Dimensions

- Training features: (212, 11)
- Testing features: (91, 11)
- Training target: (212,)
- Testing target: (91,)



Logistic Regression: Initial Results

Model Performance Metrics

75%

Accuracy

Overall correct classification rate

78%

Precision (Class 1)

When predicting disease, correct 78% of the time

76%

Recall (Class 1)

Identifies 76% of actual disease cases

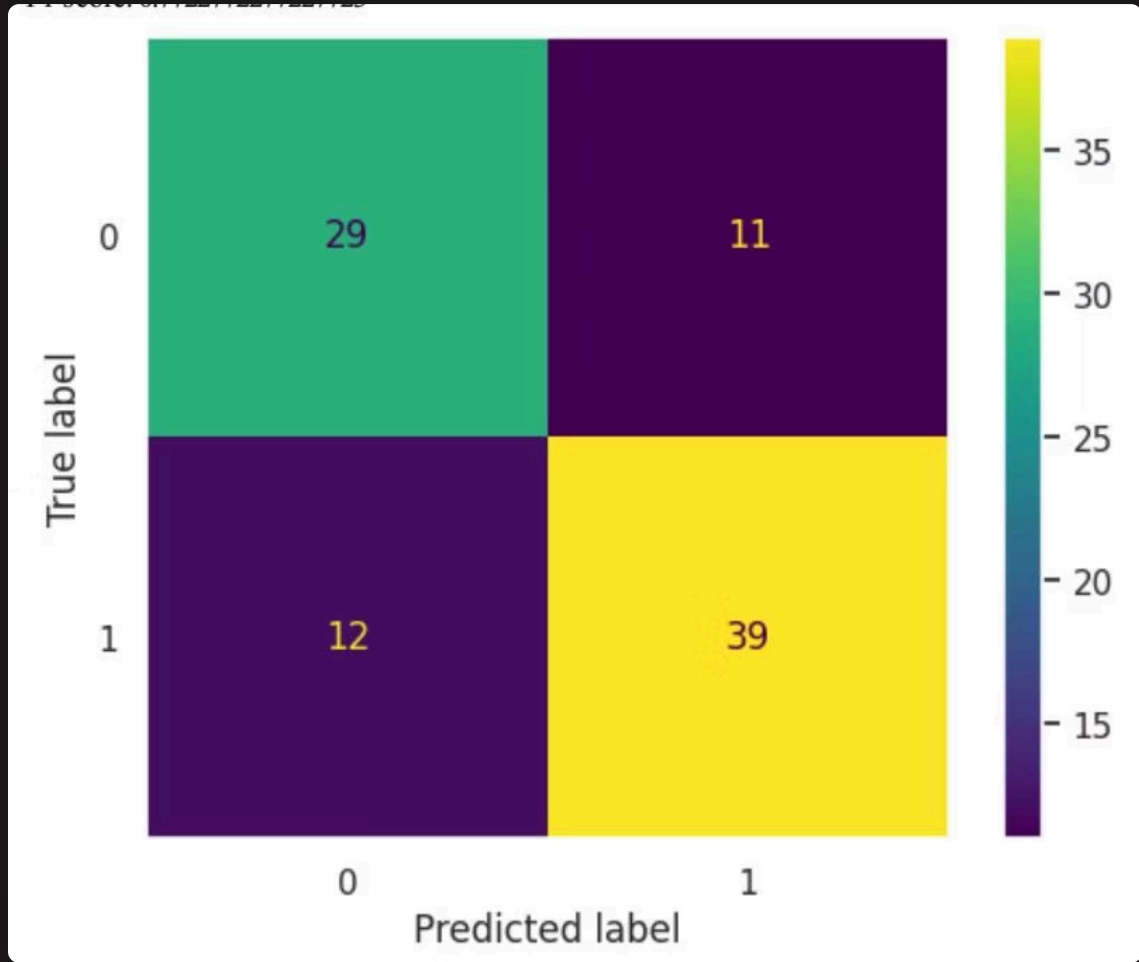
0.77

F1-Score

Good balance between precision and recall

Baseline Comparison

Dummy classifier (most frequent strategy) achieved only **56% accuracy**, confirming our logistic regression model provides substantial predictive value over random guessing.



Confusion matrix showing 29 true negatives, 11 false positives, 12 false negatives, and 39 true positives

Hyperparameter Tuning and Optimization

1

Grid Search

Tested C values: [0.001, 0.01, 0.1, 1, 10, 100] using 5-fold cross-validation

2

Best Parameter

C = 0.1 identified as optimal regularization strength

3

Improved Results

Accuracy increased from 75% to **79%** - a 4.5% improvement

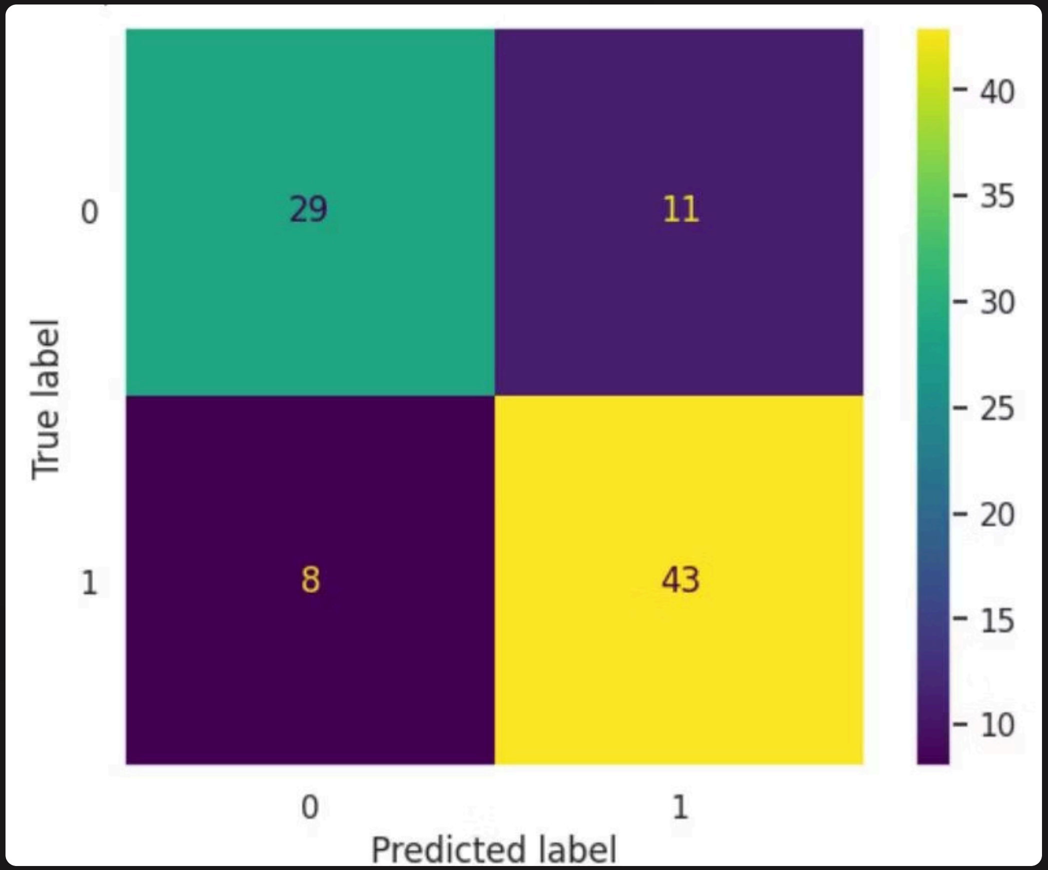
Optimized Performance

- **Accuracy:** 79.1%
- **Precision (Class 0):** 78%
- **Precision (Class 1):** 80%
- **Recall (Class 0):** 72%
- **Recall (Class 1):** 84%
- **F1-Score (Class 1):** 0.82

Cross-Validation Score

Mean Accuracy: 84.5% ± 4.8%

The low standard deviation indicates stable, consistent performance across different data splits with minimal overfitting.

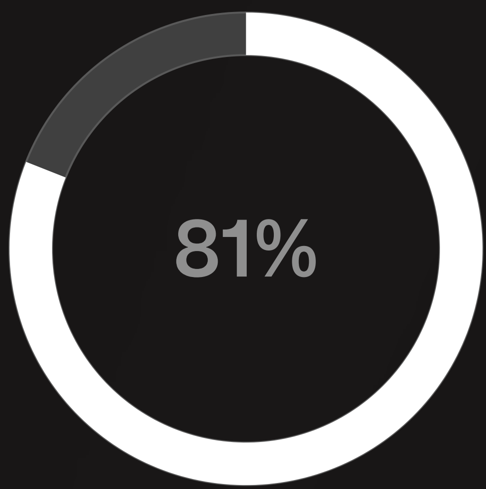


Improved confusion matrix after hyperparameter tuning

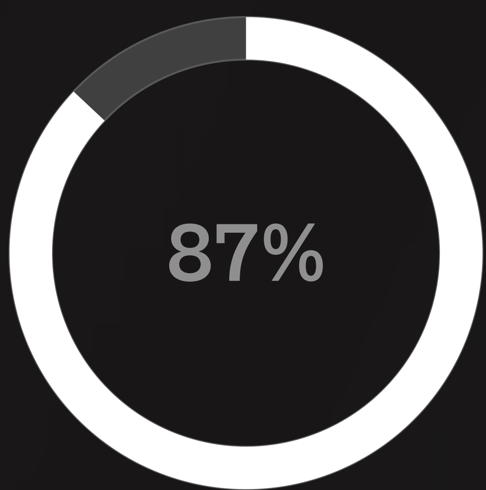
Decision Tree Model: Superior Performance

The decision tree classifier was trained as an alternative approach, with hyperparameter tuning revealing **max_depth=3** as optimal.

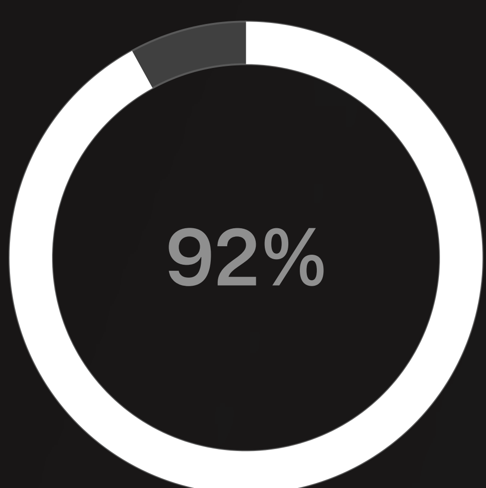
Decision Tree Metrics



Accuracy



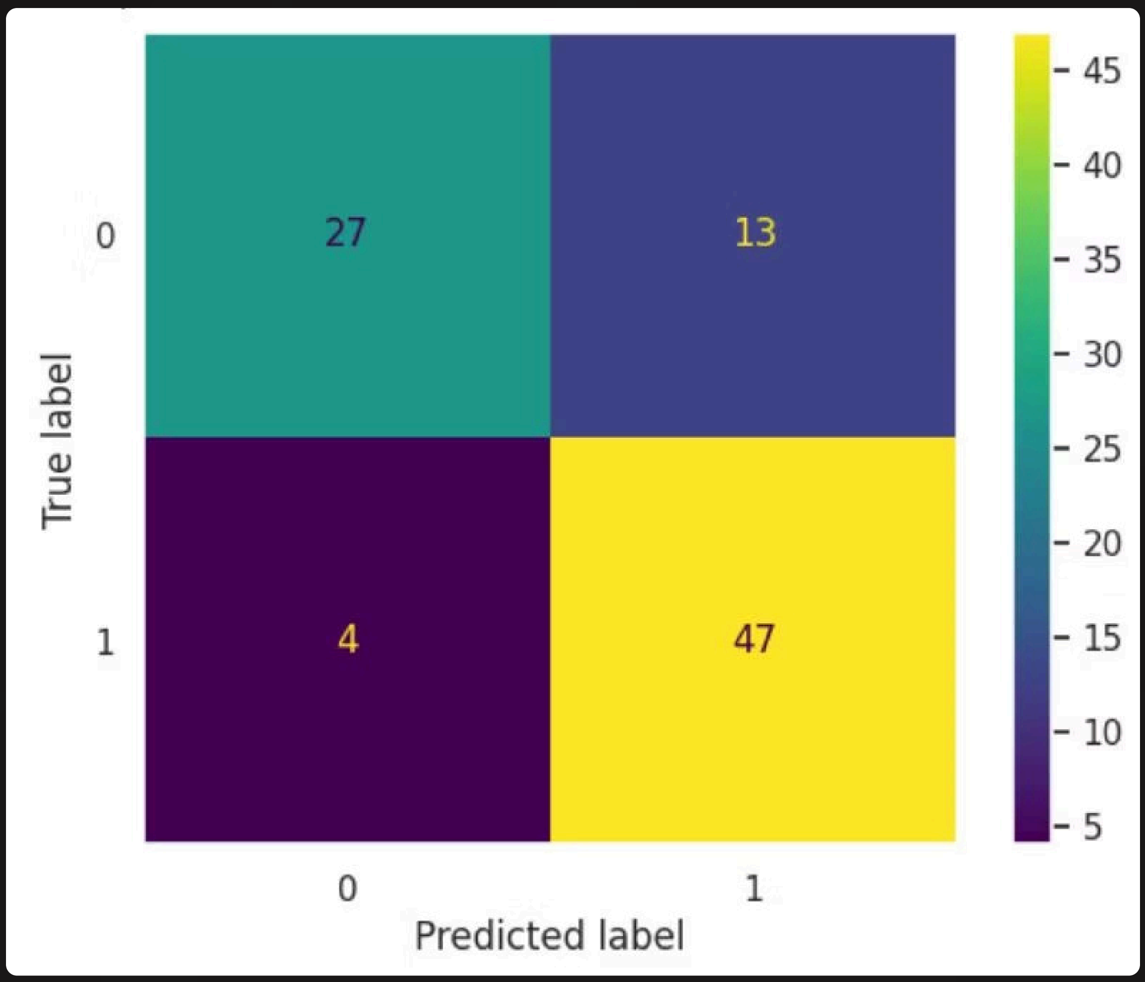
Precision (Class 0)



Recall (Class 1)

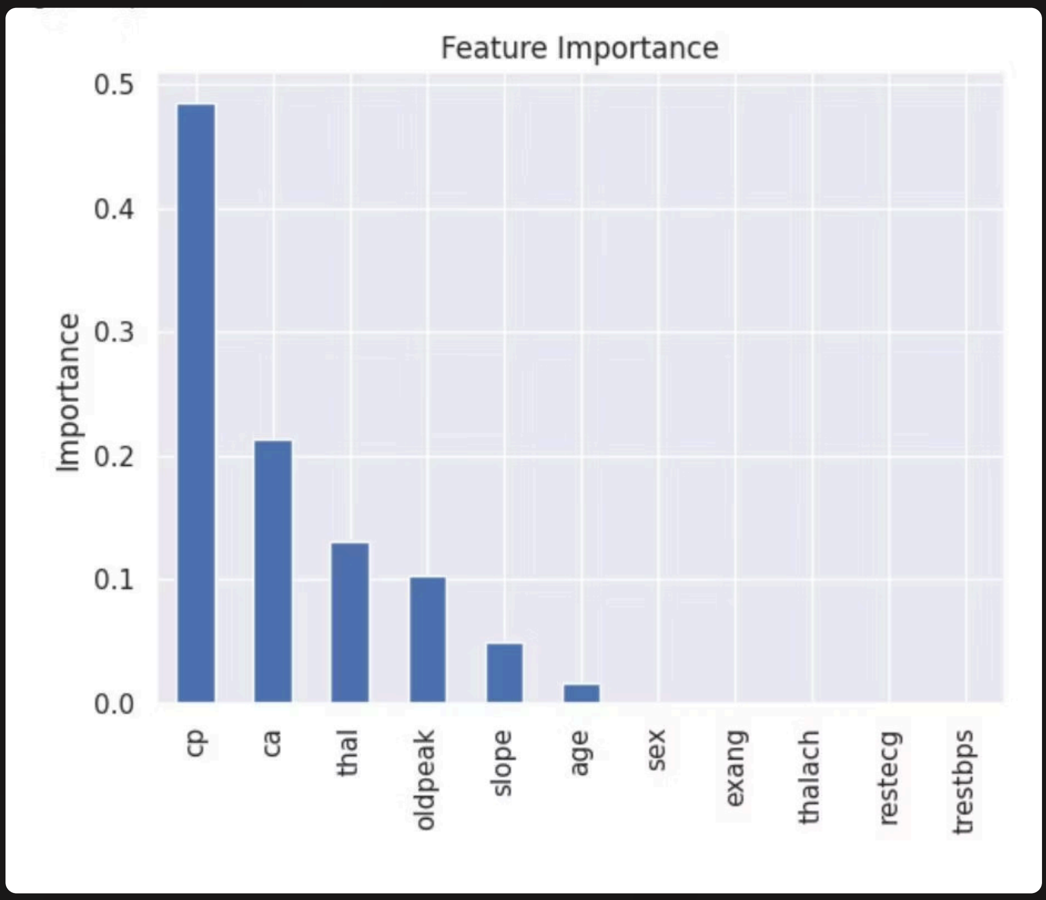
Cross-Validation: 83.0% ± 4.9%

Slightly less accurate than logistic regression in cross-validation but demonstrates excellent recall for disease detection.



Decision tree confusion matrix showing strong performance

Feature Importance Analysis



Top Contributing Features:

1. **cp (chest pain):** 48.5% importance
2. **ca (vessels):** 21.4% importance
3. **thal:** 13.1% importance
4. **oldpeak:** 10.4% importance
5. **slope:** 5.0% importance

Chest pain type emerges as the dominant predictor, contributing nearly half of the model's decision-making power.

Final Model Selection and Clinical Considerations

False Negative Risk

Model misses a positive patient, declaring them "healthy" when actually ill. **Consequence:** Patient doesn't receive timely care or treatment, potentially leading to serious health complications or death.

False Positive Risk

Model overdiagnoses, declaring patient "ill" when actually healthy. **Consequence:** Patient experiences anxiety, undergoes unnecessary tests or treatments, but is not in direct danger.

Recommended Model: Logistic Regression

In medical diagnostics, **minimizing false negatives is paramount**. While the decision tree showed slightly higher test accuracy (81% vs 79%), the logistic regression model demonstrates:

- **Superior cross-validation performance:** 84.5% vs 83.0%
- **Better generalization:** Lower risk of overfitting
- **Clinical safety:** Fewer missed disease cases
- **Interpretability:** Clear feature coefficients for medical review

The 4.5% accuracy improvement through hyperparameter tuning demonstrates the critical importance of model optimization in achieving reliable predictions.

Key Insight: In healthcare applications, maximizing recall (sensitivity) takes precedence over overall accuracy. It's better to err on the side of caution, ensuring sick patients receive necessary care even if it means some false alarms.